

In [1]:

```
#DataFrame
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv')
print(df)
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	
..	...	...	...	...	
995	female	group E	master's degree	standard	
996	male	group C	high school	free/reduced	
997	female	group C	high school	free/reduced	
998	female	group D	some college	standard	
999	female	group D	some college	free/reduced	

	test preparation course	math score	reading score	writing score
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75
..	...	...	...	...
995	completed	88	99	95
996	none	62	55	55
997	completed	59	71	65
998	completed	68	78	77
999	none	77	86	86

[1000 rows x 8 columns]

In [2]:

```
print(df[0:5])
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	

	test preparation course	math score	reading score	writing score
0	none	72	72	74
1	completed	69	90	88
2	none	90	95	93
3	none	47	57	44
4	none	76	78	75

In [65]:

```
import pandas as pd

list1 = ['New Delhi', 'Banglore', 'Mysore', 'CHD', 'Pondicherry']

df = pd.DataFrame(list1)
print(df)
```

	0
0	New Delhi
1	Banglore
2	Mysore
3	CHD
4	Pondicherry

In [5]:

```
import pandas as pd

dict = {'Name': ["Rohan", "Joe", "Raj", "Sonal", "Riya"], 'Age': [34, 53, 23, 44, 32]}

df = pd.DataFrame(dict)
print(df)
```

	Name	Age
0	Rohan	34
1	Joe	53
2	Raj	23
3	Sonal	44
4	Riya	32

In [14]:

```
df = pd.DataFrame(dict)
print(df[['Age']])
```

	Age
0	34
1	53
2	23
3	44
4	32

In [19]:

```
import pandas as pd

data = {'Name': ['Jai', 'Princi', 'Gaurav', 'Anuj'],
        'Age': [27, 24, 22, 32],
        'Address': ['Delhi', 'Kanpur', 'Allahabad', 'Kannauj'],
        'Qualification': ['Msc', 'MA', 'MCA', 'Phd']}

# Convert the dictionary into DataFrame
df = pd.DataFrame(data)

# select three columns
print(df[['Name', 'Address', 'Qualification']])
```

	Name	Address	Qualification
0	Jai	Delhi	Msc
1	Princi	Kanpur	MA
2	Gaurav	Allahabad	MCA
3	Anuj	Kannauj	Phd

In [20]:

```
result = df.loc[0]
print(result)
```

Name	Jai
Age	27
Address	Delhi
Qualification	Msc

Name: 0, dtype: object

In [21]:

```
result = df.loc[3]
print(result)
```

Name	Anuj
Age	32
Address	Kannauj
Qualification	Phd

Name: 3, dtype: object

In [27]:

```
result = df.loc[0:2, ['Name', 'Address']]
print(result)
```

	Name	Address
0	Jai	Delhi
1	Princi	Kanpur
2	Gaurav	Allahabad

In [28]:

```
result = df.loc[0:, ['Name', 'Address']]  
print(result)
```

	Name	Address
0	Jai	Delhi
1	Princi	Kanpur
2	Gaurav	Allahabad
3	Anuj	Kannauj

In [29]:

```
result = df.loc[2:, ['Name', 'Address']]  
print(result)
```

	Name	Address
2	Gaurav	Allahabad
3	Anuj	Kannauj

In [34]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv', index_col
= 'gender')
male = df.loc['male']
female = df.loc['female']
print(male)
print(female)
```

	race/ethnicity	parental level of education	lunch \
gender			
male	group A	associate's degree	free/reduced
male	group C	some college	standard
male	group B	some college	free/reduced
male	group D	high school	free/reduced
male	group C	associate's degree	standard
...	...	...	...
male	group A	high school	standard
male	group E	some high school	standard
male	group E	high school	free/reduced
male	group A	high school	standard
male	group C	high school	free/reduced

	test preparation course	math score	reading score	writing score
gender				
male	none	47	57	44
male	none	76	78	75
male	none	40	43	39
male	completed	64	64	67
male	none	58	54	52
...	...	...	...	...
male	none	57	51	54
male	completed	81	75	76
male	completed	86	81	75
male	none	63	63	62
male	none	62	55	55

[482 rows x 7 columns]

	race/ethnicity	parental level of education	lunch \
gender			
female	group B	bachelor's degree	standard
female	group C	some college	standard
female	group B	master's degree	standard
female	group B	associate's degree	standard
female	group B	some college	standard
...	...	...	...
female	group D	bachelor's degree	free/reduced
female	group E	master's degree	standard
female	group C	high school	free/reduced
female	group D	some college	standard
female	group D	some college	free/reduced

	test preparation course	math score	reading score	writing score
gender				
female	none	72	72	74
female	completed	69	90	88
female	none	90	95	93
female	none	71	83	78
female	completed	88	95	92
...	...	...	...	...
female	none	62	72	74
female	completed	88	99	95
female	completed	59	71	65
female	completed	68	78	77
female	none	77	86	86

[518 rows x 7 columns]

In [35]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv', index_col
= 'race/ethnicity')
ga = df.loc['group A']
gc = df.loc['group C']
print(ga)
print(gc)
```

race/ethnicity	gender	parental level of education	lunch \
group A	male	associate's degree	free/reduced
group A	male	some college	standard
group A	female	master's degree	standard
group A	male	master's degree	free/reduced
group A	female	associate's degree	standard
...	...	...	...
group A	female	some college	standard
group A	female	some college	standard
group A	male	high school	standard
group A	female	some high school	free/reduced
group A	male	high school	standard

race/ethnicity	test preparation course	math score	reading score \
group A	none	47	57
group A	completed	78	72
group A	none	50	53
group A	none	73	74
group A	completed	55	65
...	...	...	...
group A	none	54	63
group A	completed	78	87
group A	none	57	51
group A	none	44	45
group A	none	63	63

race/ethnicity	writing score
group A	44
group A	70
group A	58
group A	72
group A	62
...	...
group A	67
group A	91
group A	54
group A	45
group A	62

[89 rows x 7 columns]

race/ethnicity	gender	parental level of education	lunch \
group C	female	some college	standard
group C	male	some college	standard
group C	male	associate's degree	standard
group C	female	some high school	standard
group C	male	high school	standard
...	...	...	...
group C	female	associate's degree	standard
group C	female	some high school	standard
group C	female	associate's degree	standard
group C	male	high school	free/reduced
group C	female	high school	free/reduced

race/ethnicity	test preparation course	math score	reading score \
group C	completed	69	90
group C	none	76	78

group C	none	58	54
group C	none	69	75
group C	none	88	89
...	...	...	...
group C	none	91	95
group C	none	74	75
group C	none	40	59
group C	none	62	55
group C	completed	59	71

writing score

race/ethnicity

group C	88
group C	75
group C	52
group C	78
group C	86
...	...
group C	94
group C	82
group C	51
group C	55
group C	65

[319 rows x 7 columns]

In [37]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv')
print(df[0:5])
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	

	test preparation course	math score	reading score	writing score
0	none	72.0	72.0	74.0
1	NaN	NaN	90.0	88.0
2	none	90.0	NaN	93.0
3	none	47.0	57.0	44.0
4	none	76.0	78.0	NaN

In [44]:

```
import pandas
print(df[0:5].isnull())
```

	gender	race/ethnicity	parental level of education	lunch	\
0	False	False	False	False	
1	False	False	False	False	
2	False	False	False	False	
3	False	False	False	False	
4	False	False	False	False	

	test preparation course	math score	reading score	writing score
0	False	False	False	False
1	True	True	False	False
2	False	False	True	False
3	False	False	False	False
4	False	False	False	True

In [43]:

```
import pandas
print(df[0:5].fillna(0))
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
1	female	group C	some college	standard	
2	female	group B	master's degree	standard	
3	male	group A	associate's degree	free/reduced	
4	male	group C	some college	standard	

	test preparation course	math score	reading score	writing score
0	none	72.0	72.0	74.0
1	0	0.0	90.0	88.0
2	none	90.0	0.0	93.0
3	none	47.0	57.0	44.0
4	none	76.0	78.0	0.0

In [46]:

```
print(df[0:5].dropna())
```

	gender	race/ethnicity	parental level of education	lunch	\
0	female	group B	bachelor's degree	standard	
3	male	group A	associate's degree	free/reduced	

	test preparation course	math score	reading score	writing score
0	none	72.0	72.0	74.0
3	none	47.0	57.0	44.0

In [54]:

```
import pandas as pd

data = {'Name':['Jai', 'Princi', 'Gaurav', 'Anuj'],
        'Age':[27, 24, 22, 32],
        'Address':['Delhi', 'Kanpur', 'Allahabad', 'Kannauj'],
        'Qualification':['Msc', 'MA', 'MCA', 'Phd']}

# Convert the dictionary into DataFrame
df = pd.DataFrame(data)

for i, j in df.iterrows():
    print(i, j)
    print()
```

```
0 Name          Jai
Age           27
Address       Delhi
Qualification  Msc
Name: 0, dtype: object
```

```
1 Name          Princi
Age           24
Address       Kanpur
Qualification  MA
Name: 1, dtype: object
```

```
2 Name          Gaurav
Age           22
Address       Allahabad
Qualification  MCA
Name: 2, dtype: object
```

```
3 Name          Anuj
Age           32
Address       Kannauj
Qualification  Phd
Name: 3, dtype: object
```

In [52]:

```
print(df.loc[2])
```

```
Name          Gaurav
Age           22
Address       Allahabad
Qualification  MCA
Name: 2, dtype: object
```

In [61]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv', index_col
= "gender")
print(df[0:5])
```

	race/ethnicity	parental level of education	lunch	
gender				
female	group B	bachelor's degree	standard	
female	group C	some college	standard	
female	group B	master's degree	standard	
male	group A	associate's degree	free/reduced	
male	group C	some college	standard	

	test preparation course	math score	reading score	writing score
gender				
female	none	72.0	72.0	74.0
female	NaN	NaN	90.0	88.0
female	none	90.0	NaN	93.0
male	none	47.0	57.0	44.0
male	none	76.0	78.0	NaN

In [66]:

```
del list
```

In [69]:

```
import pandas as pd

data = {'Name':['Jai', 'Princi', 'Gaurav', 'Anuj'],
        'Age':[27, 24, 22, 32],
        'Address':['Delhi', 'Kanpur', 'Allahabad', 'Kannauj'],
        'Qualification':['Msc', 'MA', 'MCA', 'Phd']}

# Convert the dictionary into DataFrame
df = pd.DataFrame(data)
```

In [73]:

```
columns = list(df)
print(columns)
for i in columns:
    print (df[i][3])
```

```
['Name', 'Age', 'Address', 'Qualification']
Anuj
32
Kannauj
Phd
```

In [74]:

```
columns = list(df)
print(columns)
```

```
['Name', 'Age', 'Address', 'Qualification']
```

In [75]:

```
df.insert(3, 'Department', 'New')
columns = list(df)
print(columns)
```

```
['Name', 'Age', 'Address', 'Department', 'Qualification']
```

In [79]:

```
print(df)
```

	Name	Age	Address	Department	Qualification
0	Jai	27	Delhi	New	Msc
1	Princi	24	Kanpur	New	MA
2	Gaurav	22	Allahabad	New	MCA
3	Anuj	32	Kannauj	New	Phd

In [90]:

```
print(type(df["Qualification"][0]))
print(type(df["Department"][0]))
print((df["Department"][0]))
```

```
<class 'str'>
```

```
<class 'str'>
```

```
New
```

In [93]:

```

for i in range(len(df)):
    if df["Qualification"][i] == 'Msc':
        df["Department"][i]='Forensic'

    elif df["Qualification"][i] == 'MA':
        df["Department"][i]='Arts'

    elif df["Qualification"][i] == 'MCA':
        df["Department"][i]='UIC'

    else:
        df["Department"][i]= "New"

print(df)

```

	Name	Age	Address	Department	Qualification
0	Jai	27	Delhi	Forensic	Msc
1	Princi	24	Kanpur	Arts	MA
2	Gaurav	22	Allahabad	UIC	MCA
3	Anuj	32	Kannauj	New	Phd

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:3: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

This is separate from the ipykernel package so we can avoid doing imports until

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:6: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:9: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

if __name__ == '__main__':

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:12: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

if sys.path[0] == '':

In [123]:

```
del df
```

In [127]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\archive\\StudentsPerformance.csv', index_col
= "gender")

df.insert(7, 'total marks', '--')
df.insert(8, 'grade', '--')

df[0:5].fillna(0)

for i in range(0,5):
    df["total marks"][i] = (df["math score"][i] + df['reading score'][i]+ df['writing
score'][i])/3

for i in range(0,5):

    if df["total marks"][i] > 90.0:
        df["grade"][i]='A+'

    elif df["total marks"][i] > 80.0:
        df["grade"][i]='A'

    elif df["total marks"][i] > 60.0:
        df["grade"][i]='B+'

    elif df["total marks"][i] > 50.0:
        df["grade"][i]='B'

    else:
        df["grade"][i]= 'C'

print(df[0:5])
```

	race/ethnicity	parental level of education	lunch	
gender				
female	group B	bachelor's degree	standard	
female	group C	some college	standard	
female	group B	master's degree	standard	
male	group A	associate's degree	free/reduced	
male	group C	some college	standard	

	test preparation course	math score	reading score	writing score
gender				
female	none	72.0	72.0	74.0
female	NaN	NaN	90.0	88.0
female	none	90.0	NaN	93.0
male	none	47.0	57.0	44.0
male	none	76.0	78.0	NaN

	total marks	grade
gender		
female	72.6667	B+
female	NaN	C
female	NaN	C
male	49.3333	C
male	NaN	C

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:11: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

This is added back by InteractiveShellApp.init_path()

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:23: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

C:\Users\HP\anaconda3\lib\site-packages\ipykernel_launcher.py:29: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

In []: